

Raphael Bessin

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EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Science, John Martinson Honors College

Sep. 2025 – May 2029

- GPA: 4.00/4.00 | Dean's List: Fall 2025 - Spring 2026 | Khoury College of Computer Sciences
- Relevant Coursework: Object-Oriented Design, Algorithms & Data Structures, Discrete Structures

French American School of New York

New York, NY

High School Diploma & International French Baccalauréat

Sep. 2021 – June 2025

- Ranked 1st in grade | Mention Très Bien (Highest Honors) | High Honors Math & Physics

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C#, SQL, GDScript, HTML, CSS

Frameworks & Libraries: React, Next, Node, Express, Spring, Dash, Prisma, NumPy

Tools & Databases: Claude Code, Git, Docker, Unity, Godot, Swagger, PostgreSQL, MongoDB, Firebase, Oracle DB

Spoken Languages: English (native), French (native), German (proficient)

EXPERIENCE

FADEL

New York, NY

Software Engineering Intern

May 2026 – July 2026

- Built a Spring Boot proxy and API documentation platform, covering **9+** domains (auth, deals, royalties, financials, SFTP) with dual **JSON/XML** format switching and bearer token auth flows integrated directly in **Swagger UI**.
- Built **Claude Code** skills for endpoint calling, bulk testing, and natural language db querying, surfacing SQL injection, silent errors, and type truncations across upstream services, authoring **10+** remediation proposals.
- Implemented **10** new APIs with authentication, pagination, sorting, and custom error handling.

Northeastern Electric Racing

Boston, MA

Software Lead

Sep. 2025 – Present

- Architected a canonical-unit conversion system with global metric/imperial defaults and per-field overrides across **20+** conversions; reduced steering oscillation **80%** via a low-pass filter on Stanley controller output in NERSim.
- Implemented global car filter on FinishLine (270+ users, 100,000+ LOC), replacing duplicated logic with a single reusable filter component via middleware-backed filtering across **20+** endpoints, after the feature had stalled.
- Proposed and built FinishLine's first developer documentation system — a statically generated **Docusaurus** site auto-generating Claude AI skills from docs to avoid maintaining two separate sources of truth for 30+ developers.

PROJECTS

CurbYourCarbon | JavaScript, Chrome Extensions API, Performance API, ElectricityMaps API

2026

- Won **Social Impact & Sustainability** at HackBeanpot; built initial MVP and led team through full feature set.
- Built a CO₂ estimation engine combining network usage (Browser Performance **API**) and local carbon intensity (ElectricityMaps **API**), with personalized recommendations and goal tracking; published to Chrome Web Store.

Signal Relay | Godot, GDScript | github.com/rbessin/signal-relay

2025

- Designed a **chip abstraction system** with automatic pin detection and component generation from pin layout.
- Fixed **multi-minute freezes** caused by nested components eagerly instantiating internals; built a two-state architecture, visual-only by default with full circuit instantiated on simulate, cutting load times to **seconds**.

Stride Editor | Next.js, TypeScript, Quill.js | github.com/rbessin/stride-editor

2025

- Browser-based rich text editor powered by **n-gram language models** trained on **10** Project Gutenberg books.
- Built full data pipeline to process raw text, plus VSCode-style file/folder navigation via File System Access API.

ADDITIONAL INFORMATION

Interests: Long-Distance Running | Chess | Hiking | Nature Photography | Car Racing